



Porcelain Fused to Metal: PFM

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PFM



What is a PFM crown? These are crowns that have a metal substructure that has a ceramic veneer. These two materials are fused together with mechanical and chemical bonds.

History

- First success late 1800's
- Mid 1950's developed dental porcelains
- Became Commercially available late 1950's



If we look back into history, these types of restorations date back to the late 1800's when a porcelain jacket crown was fired onto a platinum matrix. The PFM we know today was developed in the 1950's. During this time porcelains were developed that had a similar Coefficient of Thermal Expansion to the casting metals alloying them to be fired together. By the end of the 1950's these types of crowns became commercially available to dentist.

Indications

- Patients with Esthetic Concerns
- Fixed partial Dentures
- Surveyed crowns



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When would a PFM be indicated? The indication for these types of crowns would be areas that take high stresses but the patient has esthetic concerns. Another indication would be a fixed partial denture. These types of restoration will be covered in another course, but they also undergo a lot of flexure in the framework. The casting metals are stronger than materials like zirconia and are better able to give a long term success. The last indication is for a surveyed crown. These are crowns that are designed to be under a partial denture. The metal provides a stable area to interact with the metals of a partial framework.

Contraindications

- Young patients with large pulp chambers
- Highly Esthetic areas
- When a more conservative restoration can be done



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Fig 10.22 Contemporary Fixed Prosthodontics 6th Edition 12
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As with anything there are contraindications. The first is a young patient that has a large pulp chamber. Because we need to remove more tooth structure to provide adequate thickness of the porcelain, there is a higher possibility of pulpal irritation and possible need for root canal therapy. Another contraindication are areas of high esthetics. With a metal substructure these types of restorations have a tendency to be grayish in color and have a gray hue at the margin. This can be very unesthetic to our patients. Lastly, we want to avoid these types of restorations when a more conservative restoration can be done, like an onlay.



Advantages/Disadvantages

- Get best of both worlds
- Works well in high stress areas

Vs

- Significant Reduction
- Subgingival Margins for Esthetics
- Appear Grayish
- Subject to brittle fractures

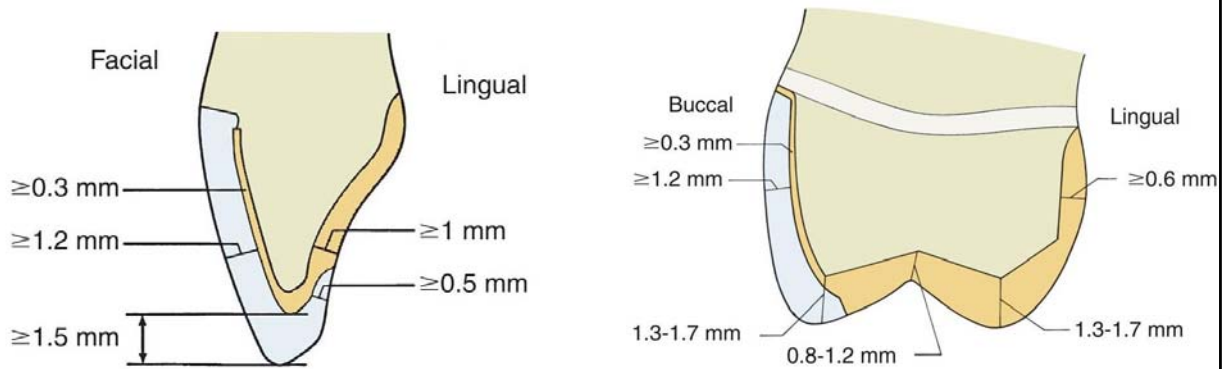
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These types of restoration give you the benefits of both a porcelain restoration and a cast metal crown. They work in high stress areas and are somewhat esthetic. Now they do require more reduction in comparison to a full cast metal, their margins will need to be subgingival to hide the grayish appearance. Lastly, they can be subject to brittle fractures.

Prep Dimensions



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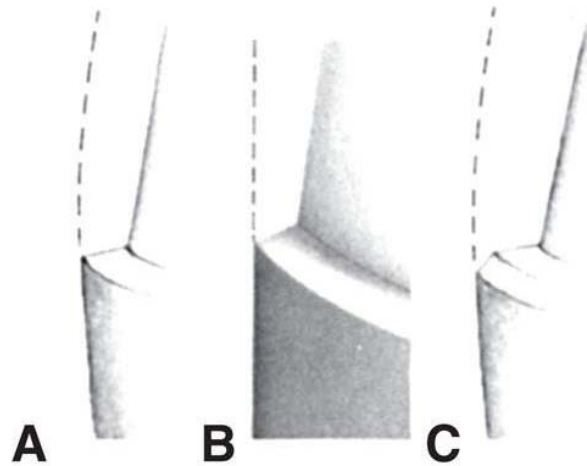
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Fig 10.1 Contemporary Fixed Prosthodontics 6th Edition

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Figure 10.1 from Contemporary Fixed Prosthodontics 6th edition gives prep dimensions from these types of restorations. Each of the examples given show only porcelain on the Facial or Buccal aspect of the crown. This will need to be taken into account depending on what design you would like the crown to have. It is important to communicate with your lab technician so they understand what you would like.

Marginal Geometry



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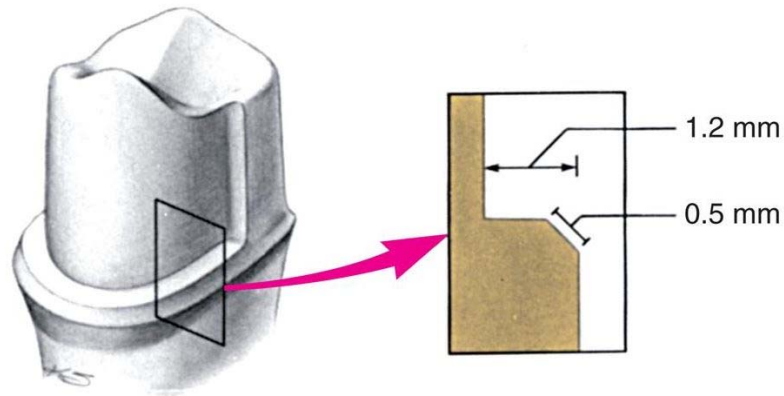
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Fig 10.16 Contemporary Fixed Prosthodontics 6th Edition 12
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The marginal geometry for a PFM will be dictated by the margin that you as the practitioner has prescribed. If you are going to have a lingual in metal, then a chamfer is needed. If you want a porcelain margin, then a shoulder or beveled shoulder are the appropriate marginal geometries.

Due to the brittle nature of porcelain, we ideally want a cavosurface angle of 90 degrees. But the prevalence of unsupported enamel at the finish line dictates that either an obtuse cavosurface angle of 135 degrees or a beveled shoulder is more appropriate and will give longer marginal longevity.

Marginal Designs: Beveled Shoulder



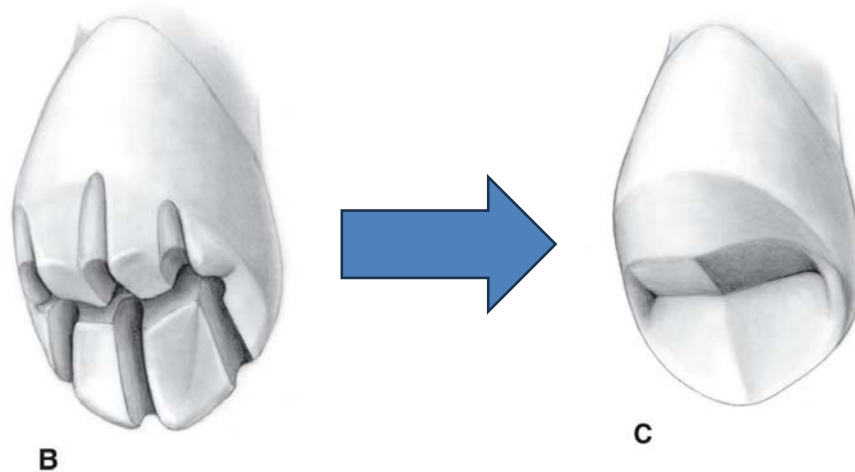
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Fig 10.17 Contemporary Fixed Prosthodontics 6th Edition 13
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When doing a beveled shoulder, the recommendation is to have a short bevel at the finish line. These types of margins will then have the casting metal at the finish line creating a better seal. Additionally, the metal will provide support for the porcelain. Unfortunately, in the anterior these types of marginal geometry are not very esthetic and will require a subgingival margin placement to hide the metal.

Prep Technique



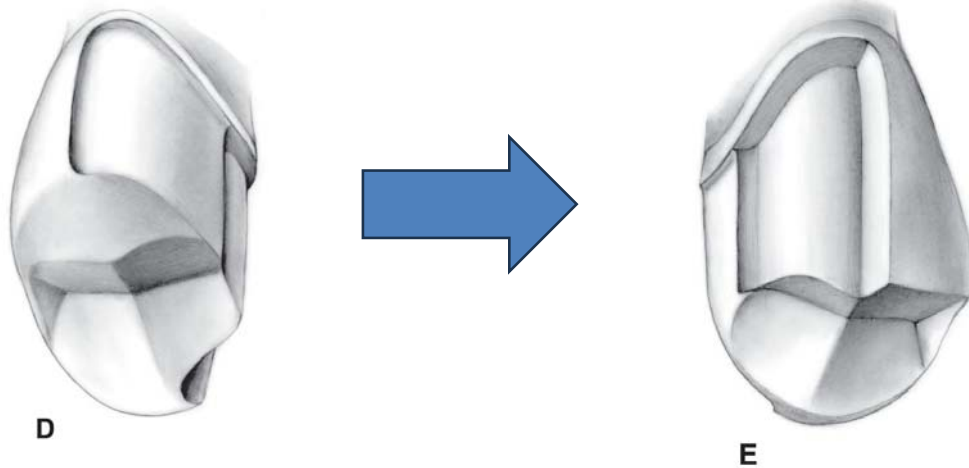
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Fig 10.3 Contemporary Fixed Prosthodontics 6th Edition 13
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Preparing a PFM is very similar if not exactly the same as an all porcelain crown. The example from the textbook shows occlusal first reduction then axial. This can be flipped according to the provider preferred technique.

Prep Technique



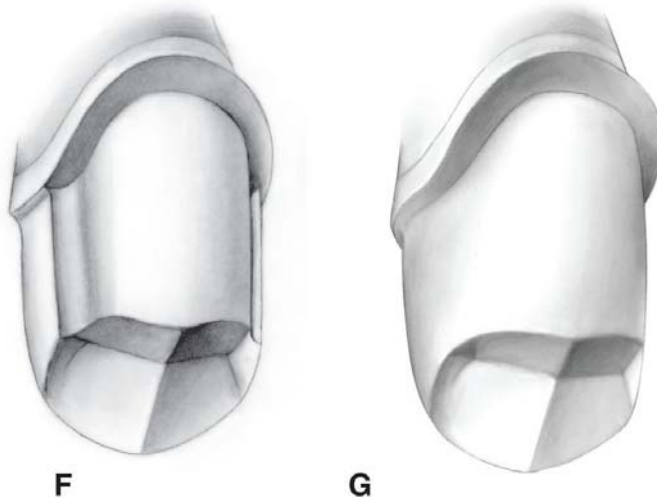
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Fig 10.3 Contemporary Fixed Prosthodontics 6th Edition 13
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With many PFM crowns, the lingual will have a metal component especially at the margin. When this occurs, we will need to have differing marginal depths. This is demonstrated in the two images. The buccal portion that will have a porcelain margin will be deeper than the lingual with a metal margin. Now, this is not mandatory. You as the provide will decide the design. If you want porcelain all the way around, then you will need to prep the tooth for this and communicate to your lab technician your design.

Prep Technique



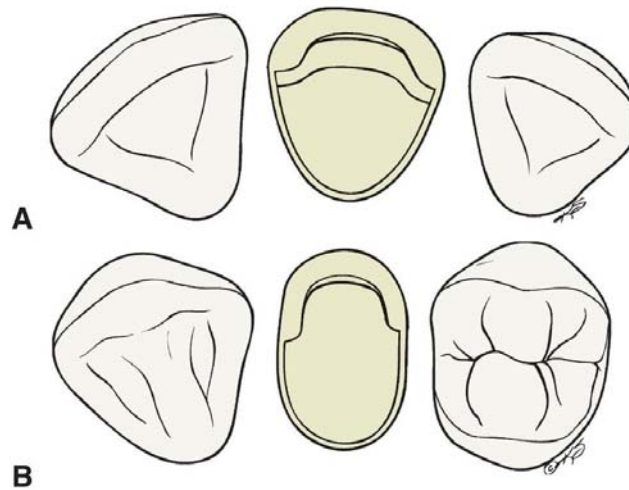
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Fig 10.3 Contemporary Fixed Prosthodontics 6th Edition 13
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The variable depth preparation can be left 2 different ways, with or without wings. Figure F shows what it looks like to leave wings in your preparation. This can cause some stress in the restoration but can help to provide extra retention on a tooth that has a short clinical height. Another technique is to remove the wings and make the transition from shoulder to chamfer a smooth one. Both of these designs are perfectly acceptable and you as the clinician will need to decide what works best for you.

Interproximal Extension



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Fig 10.8 Contemporary Fixed Prosthodontics 6th Edition

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One big question that should be asked is how far should the shoulder margin go interproximally before it transitions into a chamfer. The shoulder will need to extend beyond the line angle and end before the contact or after the contact space. We do not want the porcelain metal transition to be in the contact space. Additionally, depending on where in the mouth you are working, you will need to extend the shoulder deeper to prevent any metal from being visible.

Metal Framework



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Fig 24.6 Contemporary Fixed Prosthodontics 6th Edition 13
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The metal collar of framework is the substructure that will support your porcelain. The original PFM crowns had metal margins and occlusion and only porcelain in the visible areas. This has changed over time to allow for porcelain to cover all of the surfaces of the crown. The shape of the collar should be convex with rounded contours. It should be designed so that it will support the porcelain without stress concentrations.

Metal Framework

- Noble Metal Alloys: 0.3mm
- Base Metal Alloys: 0.2mm



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Fig 24.7 Contemporary Fixed Prosthodontics 6th Edition 13
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The metal collar will need a minimal thickness so that it will not distort during the firing process. For noble metals that minimal thickness is 0.3mm. Base metals, due to their high fusing ranges, good moduli of elasticity and yield strengths, only need 0.2mm as a minimal thickness.

Metals Used

- High Noble
- Noble
- Titanium
- Base Metals



There are 4 general types of metals using as coping for a PFM crown. These are high noble metals, noble metals, titanium and base metals. Titanium became a good alternative in the late 2000's before the advent of emax and all zirconia crowns. It is not used very much anymore.

The requirements for a metal to be used are:

- 1) High melting temperature that is substantially higher than a firing temperature (at least >100 degree C)
- 2) Forms a good bond between the metal and ceramic
- 3) Coefficient of Thermal Expansion must be similar to the ceramic
- 4) Adequate stiffness and strength
- 5) High sag resistance
- 6) Accurate Casting of the metal

These metals generally are not used in a pure form. Alloys of metals are made to maximize their material properties.

Noble/Base Metals

Properties of Elements in Dental Casting Alloys							
Element	Symbol	Atomic Number	Atomic Mass	Density (g/cm ³)	Melting Temperature (°C)	Color	Comments
Noble							
Ruthenium	Ru	44	101.07	12.48	2310.0	White	Grain refiner, hard
Rhodium	Rh	45	102.91	12.41	1966.0	Silver-white	Grain refiner, soft, ductile
Palladium	Pd	46	106.42	12.02	1554.0	White	Hard, malleable, ductile
Osmium	Os	76	190.20	22.61	3045.0	Bluish-white	Not used in dentistry
Iridium	Ir	77	192.22	22.65	2410.0	Silver-white	Grain refiner, very hard
Platinum	Pt	78	195.08	21.45	1772.0	Bluish-white	Tough, ductile, malleable
Gold	Au	79	196.97	19.32	1064.4	Yellow	Ductile, malleable, soft, conductive
Base							
Nickel	Ni	28	58.69	8.91	1453.0	White	Hard
Copper	Cu	29	63.55	8.92	1083.4	Reddish	Malleable, ductile, conductive
Zinc	Zn	30	65.39	7.14	419.6	Bluish-white	Soft, brittle, oxidizes
Gallium	Ga	31	69.72	5.91	29.8	Grayish-white	Low melting
Silver	Ag	47	107.87	10.49	961.9		Soft, malleable, ductile, conductive
Tin	Sn	50	118.71	7.29	232.0	White	Soft
Indium	In	49	114.82	7.31	156.6	Gray-white	Soft

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Table 10.6 Craig's Restorative Dental Materials 14th Edition 13
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Table 10.6 from Craig's Restorative dental materials 14th edition gives examples of noble and base metals. As you can see on the right side the color of the metal is important since the patient may see this.

Noble Metals



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A noble metal is a chemical element that is generally resistant to corrosion and is found in nature in its raw form. Noble metals also resist oxidation, tarnish and corrosion while being heated, casted, soldered or used in the mouth. Gold and palladium generally form a foundation for which others are added.



Metal Requirements

Revised American Dental Association Classification of Prosthodontic Alloys

Class	Required Noble Content (%)	Required Gold Content (%)	Required Titanium Content (%)
High-noble alloys	≥60	≥40	
Titanium and titanium alloys			≥85
Noble alloys	≥25		
Predominantly base materials	≥25		

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Table 10.3 Craig's Restorative Dental Materials 14th Edition 14
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Table 10.3 reviews the requirements to call a prosthodontic alloy a certain category.

High Noble Metals

Revised American Dental Association Classification of Prosthodontic Alloys

Class	Required Noble Content (%)	Required Gold Content (%)	Required Titanium Content (%)
High-noble alloys	≥60	≥40	
Titanium and titanium alloys			≥85
Noble alloys	≥25		
Predominantly base materials	≥25		

Class	Required Noble Content (%)	Required Gold Content (%)
High-noble alloys	≥60	≥40

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Table 10.3 Craig's Restorative Dental Materials 14th Edition 14
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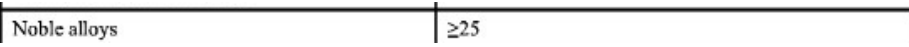
To be a high noble metal it requires that more than 60% by weight is a noble metal and at least 40% by weight is gold. As you can guess these alloys are going to be more expensive due to the gold content.

Noble Metals

Revised American Dental Association Classification of Prosthodontic Alloys

Class	Required Noble Content (%)	Required Gold Content (%)	Required Titanium Content (%)
High-noble alloys	≥60	≥40	
Titanium and titanium alloys			≥85
Noble alloys	≥25		
Predominantly base materials	≥25		

Required Noble Content (%)



A noble alloy is an alloy that has at least 25% by weight of noble metals. The difference between a noble and high noble alloy is that a noble alloy is not required to have a gold content. This makes these alloys lower cost.

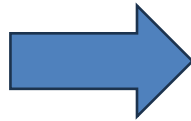
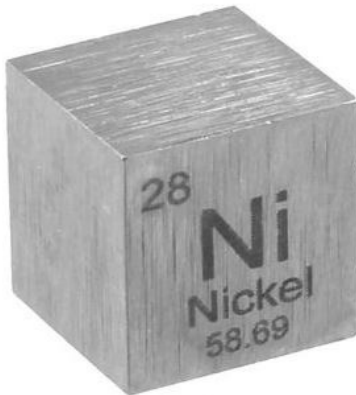
Base Metals

Composition Ranges of Base Metals for Ceramic-Metal Restorations (wt%)

Type	Ni	Cr	Co	Ti	Mo	Al	V	Fe	Be	Ga	Mn	Nb	W	B	Ru
Ni-Cr	69-77	13-16	—	—	4-14	0-4	—	0-1	0-2	0-2	0-1	—	—	—	—
Co-Cr	—	15-25	55-58	—	0-4	0-2	—	0-1	—	0-7	—	0-3	0-5	0-1	0-6
Ti	—	—	—	90-100	—	0-6	0-4	0-0.3	—	—	—	—	—	—	—

Base metal alloys common in dental castings are a nickel-chromium alloy. As you can see from the table these alloys have a high concentration of nickel by weight. Titanium as a casting metal falls into this category. As you can see these alloys have a really high concentration by weight of titanium and very little of other metals.

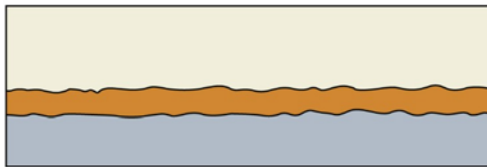
Base Metals



One major issue that can occur with base metal alloys is a reaction to nickel. Many people have an allergy to nickel. The nickel in these crowns can cause contact dermatitis in the oral cavity. This will need to be taken into account as you choose the type of metal that will be used by your laboratory technician.

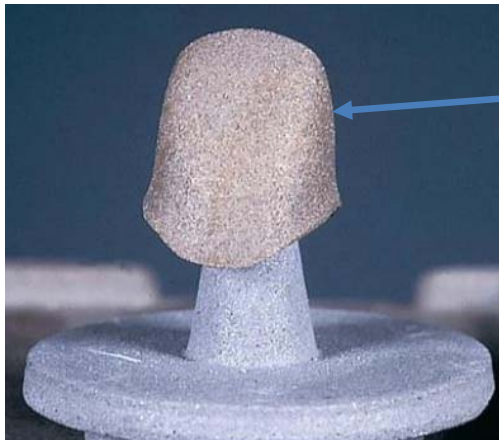
Fusion

- Requires the creation of an oxide layer
- Noble Metals
- Base Metals



Ceramic
 Oxide
 Metal

Fig 11.1 Craig's Restorative Dental Materials 14th Edition



Oxidized Layer

Fig 24.6 Contemporary Fixed Prosthodontics 6th Edition

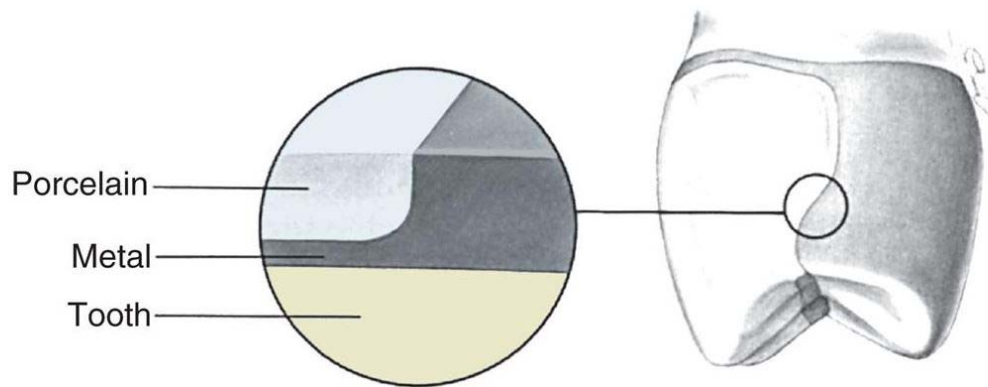
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In order to fuse the porcelain to the metal an oxide layer needs to be formed on the metal coping. This will allow for a chemical bond between the metal and the porcelain. To create this layer the metal is placed in an oven and heated in a vacuum. The vacuum is used to limit the thickness of the oxide layer. It is important to know that each alloy will form an oxide layer at different rates which will dictate the time that you heat the collar. In alloys with high gold content iron is used to create an oxide layer. In alloys with lower gold content; tin and indium are used. Alloys that have high palladium content, the gallium in the alloy will create the oxide layer. Now with base metals the chromium oxide provides for the chemical bond.

Metal Ceramic Junction



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Fig 24.4 Contemporary Fixed Prosthodontics 6th Edition 14
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The junction between metal and ceramic needs to be 90 degrees and it should be smooth. This will reduce the stress on these areas. The junction should be 1.5mm away from an occlusal centric contact. Additionally, the junction should be distinct so that the excess porcelain can easily be removed.

Porcelain Veneer



The porcelain veneer on a PFM crown is what makes it an esthetic restoration. The metal surface that will be veneered needs to be smooth in texture, have rounded line angles and allow for good wetting of the opaque porcelain. We will go into more detail of this process.

Porcelain



A

FIG. 24.19 (A to C) Types of metal-ceramic porcelain. Porcelains are available as powders or pastes. (Courtesy Ivoclar AG, Amherst, NY.)



B



C

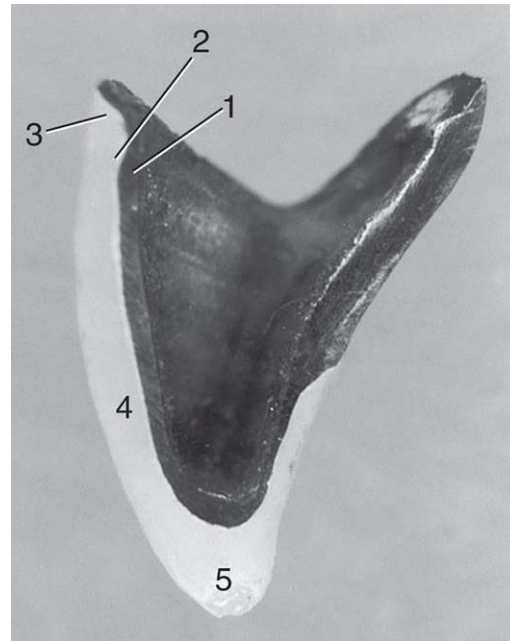
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Fig 24.19 Contemporary Fixed Prosthodontics 6th Edition 14
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Dental porcelains are made up of quartz, Feldspar (potassium aluminum silicate orthoclase, sodium aluminum silicate albite and Al_2O_3) and other oxides. These materials come in powders with grain sizes of 5-110 microns for fine grains and up to 200 microns for coarse grains. These powders are mixed in distilled water or aqueous liquids to make a slurry that is added to the metal.

Veneer Layers



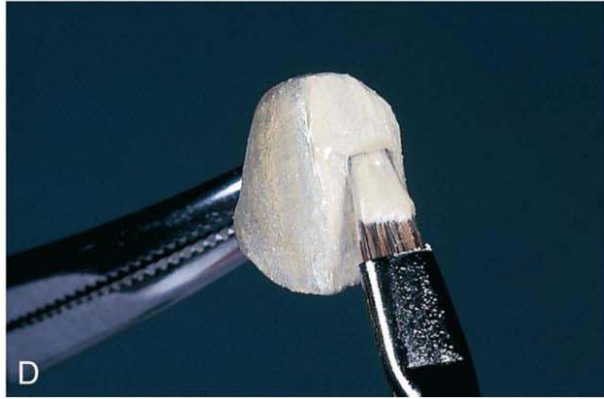
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Fig 24.2 Contemporary Fixed Prosthodontics 6th Edition 14
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Figure 24.2 from Contemporary Fixed Prosthodontics 6th edition shows the different layers of a PFM. The first layer is the metal collar. This is covered with an opaque porcelain. Layers 3 and 4 are similar with #3 being the gingival and 4 the body porcelain. Area 5 is the incisal porcelain. These layers represent portions of the crown that will have differing levels of translucency and color.

Opaque Layer



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Fig 24.21 Contemporary Fixed Prosthodontics 6th Edition 15
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The opaque layer of porcelain is used to block out the metal coloration of the collar. This layer will form the metal-ceramic bond for the porcelain. The porcelain should wet easily and should only be 0.1mm thick to achieve appropriate esthetics of the crown.

Body Porcelain



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Fig 24.23 Contemporary Fixed Prosthodontics 6th Edition 15
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The body porcelain will have some translucency and will create the “color” of the crown. Additionally, characterization of the tooth will be added in these layers. An example of this would be the mamelons seen figure E.

Incisal Porcelain



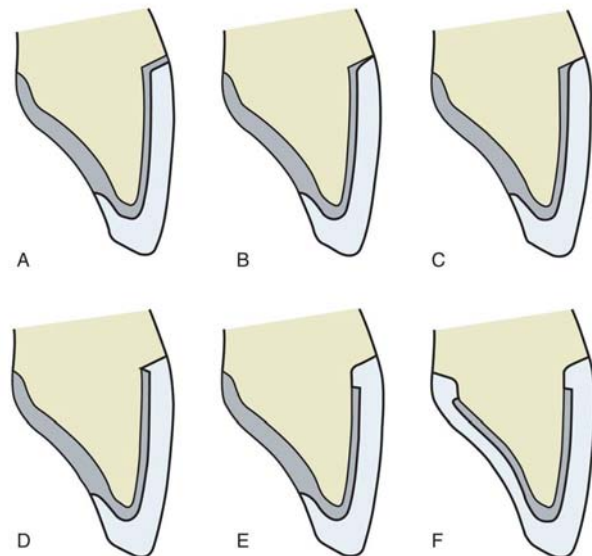
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Fig 24.26 Contemporary Fixed Prosthodontics 6th Edition 15
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The incisal porcelain will have the most translucency allowing you to see the forms developed in the body. Following this layer a glaze will be added to smooth crown and give it the proper surface texture.

Marginal Designs



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Fig 24.32 Contemporary Fixed Prosthodontics 6th Edition 15
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There can be many marginal designs used for a facial porcelain margin. Figure 24.32 from Contemporary Fixed Prosthodontics shows the variable designs. When would we use a collar-less margin as seen in D-F. You will want to consider this when a gray marginal area is a concern. These can be advantageous in that they are esthetic, easier for plaque removal and can be placed at a crestal or supracrestal marginal position. There are few disadvantages as well. These are more difficult to fabricate, the marginal adaptation is inferior to the metal and they can fracture during try-in. All of these factors should be taken into account when making a decision on how you want to design the margin. These types of margins are contraindicated in areas that you can not achieve a depth of 1mm.

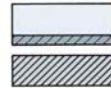
Failure

ADHESIVE FAILURE

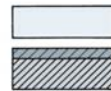
1. Porcelain-metal interface



2. Metal oxide-metal



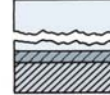
3. Porcelain-metal oxide



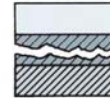
Porcelain Metal Metal oxide

COHESIVE FAILURE

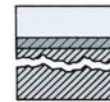
4. Porcelain-porcelain



5. Metal oxide-metal oxide

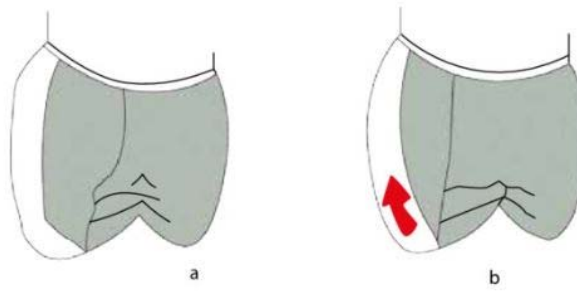


6. Metal-metal



There are 2 types of failure categories in these crowns: Adhesive and Cohesive. An adhesive failure is a failure at the point of bonding. A cohesive failure is one that occurs inside the material. This could be the porcelain, the metal oxide or the metal itself.

Unsupported Porcelain



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Fig 5 Ceramic Fracture in Metal-Ceramic Restorations:
The Aetiology, Dental Update May 2017; Aslam et al

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A common example of a cohesion failure in a PFM crown is when you have unsupported porcelain. In these instances, the metal coping was not designed so that the porcelain veneer has sound support evenly. This support should create a situation where the crown has 1-1.5mm of porcelain. Additionally, your coping design should distribute the forces so that the porcelain is in compression. Areas of porcelain that exceed 2mm in thickness has a tendency to fracture. Let look at an example of this in the image. Figure A shows even support of the porcelain veneer, where as, figure B does not and will put the buccal porcelain under shear stress.